Tien-Chen Lin

Cytoskeletal neurobiology · Mechanobiology of brain development

German Center for Neurodegenerative Diseases (DZNE), Venusberg-Campus 1, Building 99
53127 Bonn, Germany
tien-chen.lin@dzne.de
tien-chen-lin.github.io
ORCID: 0000-0002-6852-2900 *Google Scholar: xARPM5QAAAAJ*

Research summary

My research lies at the intersection of molecular neurobiology, cytoskeletal dynamics, and mechanotransduction. I study how diverse cell types coordinate their morphological changes during brain development, combining theoretical modelling of cytoskeletal dynamics with live imaging, optogenetic manipulation *in vitro* and *in vivo*, biochemical reconstitution, and large-scale omics. My long-term goal is to uncover the biophysical principles governing the self-organization of the nervous system, and their dysregulation in aging and degeneration.

Education

2009–2014 **Ph.D., Natural Sciences (Dr. rer. nat., *magna cum laude*)**, *University of Heidelberg*, Germany

2005–2006 **M.Sc., Biochemistry & Molecular Biology**, *National Yang-Ming University*, Taiwan

2002–2005 **B.Sc., Life Sciences**, *National Yang-Ming University*, Taiwan

Research experience

2017–present **Postdoctoral Researcher**, *DZNE, Bonn — Axon Growth & Regeneration (Prof. F. Bradke)*
Discovered a soma-based, Arp2/3–actomyosin local-excitation/global-inhibition mechanism of CNS neuronal polarization; reconciled the roles of actin networks in neurite growth; built an R/ImageJ analysis pipeline for neurite-growth dynamics; collaborative omics of axon regeneration, reactive astrocytes after stroke, and DRG-neuron subtypes.

2014–2017 **Postdoctoral Researcher**, *ZMBH, University of Heidelberg (Prof. E. Schiebel)*
Established MOZART1 as an integral part of the γ-tubulin small complex in *Candida albicans* and human cells, and as a driver of its assembly into a nucleation-competent ring; mapped conserved interactions among MOZART1, γ-TuSC, and the receptors pericentrin and CDK5RAP2.

2009–2014 **Doctoral Researcher**, *ZMBH, University of Heidelberg (Prof. E. Schiebel)*
Elucidated the mitotic phospho-regulation of the γ-tubulin small complex and pericentrin; characterized γ-TuSC–receptor interactions and the divergent targeting of γ-tubulin complexes to microtubule organizing centers.

Honors & fellowships

2016 **EMBO Short-term Fellowship**

2012 **LGFG Fellowship — "Regulation of Cell Division"**

Selected publications

2026 **Lin, T.-C.**, Coles, C.H., Alfadil, E., *et al.*, Bradke, F. An intrinsic cytoskeletal oscillator establishes neuronal polarity. *Nature* (2026). doi:10.1038/s41586-026-10755-6

2016 **Lin, T.-C.**, Neuner, A., Flemming, D., *et al.* MOZART1 and γ-tubulin complex receptors are both required to turn γ-TuSC into an active microtubule nucleation template. *J. Cell Biol.* 215(6), 823–840.

2015 **Lin, T.-C.**, Neuner, A., Schiebel, E. Targeting of γ-tubulin complexes to microtubule organizing centers: conservation and divergence. *Trends Cell Biol.* 25(5), 296–307.

2014 **Lin, T.-C.**, Neuner, A., Schlosser, Y.T., *et al.* Cell-cycle dependent phosphorylation of yeast pericentrin regulates γ-TuSC-mediated microtubule nucleation. *eLife* 3, e02208.

2011 **Lin, T.-C.**, Gombos, L., Neuner, A., *et al.* Phosphorylation of the yeast γ-tubulin Tub4 regulates microtubule function. *PLoS ONE* 6(5), e19700.

Full list: tien-chen-lin.github.io/publications

Invited talks (selected)

2023 **Microtubules in Neurons (Chiemsee); The Cytoskeleton of Neurons and Glia (webinar series)**

2022 **CSHL "Molecular Mechanisms of Neuronal Connectivity" (New York); FENS Symposium (Paris); EMBO Workshop (Taipei); invited lecture with F. Bradke, NYCU College of Life Sciences (Taiwan)**

Teaching & service

2020–2023 **Coordinator & lecturer**, *M.Sc. "Molecular Cell Biology", University of Bonn*
Light-microscopy basics, live-cell imaging, and data visualization with R.

2019–present **Associate Faculty Member**, *F1000 / Faculty Opinions*

Mentoring Supervision of M.Sc. research projects and rotation students.

Professional development

EMBO Laboratory Leadership for Group Leaders; Project Management for Scientists; Negotiation for Scientists.

Courses Advanced Techniques in Molecular Neuroscience (CSHL); Super-Resolution Microscopy and Transcriptome/Proteome Integration (EMBL).